

heuristic

Vincent Wang

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1 Heuristics

Let h_1, h_2 be admissible. Then $\max\{h_1(v), h_2(v)\} \leq \delta(v, T)$, so the maximum of two admissible heuristics is admissible.

Let h_1, h_2 be consistent. I claim that $\max\{h_1, h_2\}$ is consistent.

Proof: Let $u, v \in V$ be adjacent. WLOG $h_1(u) \geq h_2(u)$. Then

$$\max\{h_1(u), h_2(u)\} - \max\{h_1(v), h_2(v)\} \leq h_1(u) - h_1(v) \leq w(u, v).$$

2 Probability

$X \perp Y \mid Z$ if $p(X, Y \mid Z) = p(X \mid Z)p(Y \mid Z)$, or $p(X, Y, Z)/p(Z) = p(X, Z)/p(Z) \cdot p(Y, Z)/p(Z) \rightarrow p(X, Y, Z)p(Z) = p(X, Z)p(Y, Z)$.

$$p(X \mid Y, Z) = p(X, Y, Z)/p(Y, Z) = p(X, Z)/p(Z) = p(X \mid Z).$$

$$p(X, Y, Z) \stackrel{?}{=} p(X \mid Y, Z)p(Y \mid Z)p(Z)?$$

$$RHS = \frac{p(X, Y, Z)}{p(Y, Z)} \cdot \frac{p(Y, Z)}{p(Z)} \cdot p(Z) = p(X, Y, Z).$$

Consider $Z \rightarrow Y \rightarrow X$.

$$p(X, Y, Z)p(Y) =$$